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An Optimized Lightweight Crop Pest Detection Method

Zekai Cheng[1], Rongqing Huang[1,*], Rong Qian[2], Shu Zhang[1], Xinzhuo Jiang[1], Meifang Liu[1] and Xiwen Qu[1]

[1] School of Computer Science and Technology, Anhui University of Technology, Ma'anshan 243032, China

[2] Institute of Agricultural Economy and Information, Anhui Academy of Agricultural Sciences, Hefei 230031, China

[*] Corresponding author's e-mail: tty0013@ahut.edu.cn

Abstract. **The practical application of crop pest detection methods has been limited by the large number of parameters and computations, and we built a lightweight crop pest detection method YOLOLite-CSG in our previous research, which basically removed this limitation. However, further analysis shows that YOLOLite-CSG still has problems that affect the performance in terms of the prior box generation method, downsampling method and bounding box regression loss function. In response to these problems, this paper proposes the prior box generation method based on label box oversampling and clustering (BS-Medians), the downsampling method based on parallel feature transformation (PSPD-Conv), and the bounding box regression loss function with size difference optimization capability (FIoU Loss). We optimize YOLOLite-CSG using the above methods to build YOLOLite-X, an optimized lightweight crop pest detection method. The experiment results show that YOLOLite-X exceeds YOLOLite-CSG in crop pest detection precision and is higher than other state-of-the-art methods. Meanwhile, YOLOLite-X has a smaller number of parameters and computations, which is more conducive to practical deployment and application.**

Keywords-crop pest detection; prior box generation method; downsampling method; bounding box regression loss function

I. Introduction

Crop pest detection is the key to achieving precision agriculture. Deep learning-based two-stage object detection methods such as Faster R-CNN [1] are widely used for crop pest detection. Most research would first improve these two-stage object detection methods used in general-purpose domains to make them applicable to crop pest detection tasks. Common improvements include modifying the backbone network [2], improving non-maximal suppression methods [3], applying data augmentation methods [4], and extracting features such as grayscale covariance matrix and contrast [5].

Meanwhile, one-stage object detection methods such as YOLOv3 [6], which have a more simple and efficient structure, have gradually replaced two-stage object detection methods as the mainstream methods for crop pest detection research. One-stage object detection methods have less structural coupling and are capable of structural scaling to build detection methods of different complexity. Scalable methods such as YOLOv5 are frequently used to detect crop pests and can provide crop pest detection capabilities at different precision levels [7]. These methods are also usually modified appropriately to enhance their performance, such as modifications to the feature pyramid structure [8]. Data augmentation still plays an important role in one-stage crop pest detection methods [9], which can expand the dataset and provide sufficient data for training. The trained models are generally deployed in platforms such as Raspberry Pi to perform crop pest detection [10].

The above deep learning-based object detection methods achieve high crop pest detection precision. However, the larger number of parameters and computations limits the practical deployment and application of these methods. To address this problem, we proposed the lightweight crop pest detection method YOLOLite-CSG [11] in our previous research. YOLOLite-CSG has smaller number of parameters and computations, and higher detection precision. In order to further improve the detection precision, an in-depth analysis of YOLOLite-CSG is conducted in this paper, and some points to be optimized in the prior box generation method, downsampling method and bounding box regression loss function are found. Therefore, this paper proposes several optimization methods to improve YOLOLite-CSG and build an optimized lightweight crop pest detection method YOLOLite-X. The main works are as follows:

(1) For the imbalance of the label box distribution on the prior box generation, we propose BS-Medians, a prior box generation method based on label box oversampling and clustering.

(2) Aiming at the loss of fine-grained feature information due to strided convolution, we design PSPD-Conv, a downsampling method based on parallel feature transformation.

(3) For the function degradation in the bounding box regression process, we construct FIoU Loss, a bounding box regression loss function with size difference optimization capability.

(4) With the above methods, we build an optimized lightweight crop pest detection method, YOLOLite-X.

II. Methods

A. Optimization of prior box generation method

In the previous research, we proposed YOLOLite-CSG, which uses K-Means++ to generate the prior boxes. This process ignores the imbalance problem of the label boxes, which is prone to generate suboptimal prior boxes. The imbalance problem of the label boxes is mainly reflected in two aspects, i.e., the quantity distribution among classes and the size distribution. In this paper, the CP15 dataset is used, and the distribution of the label box quantity among the classes is shown in Figure 1, and the distribution of the label box size is shown in Figure 2. From Figure 1, the distribution of the label box quantity among the classes shows an obvious imbalance, and the label box quantity of small pests such as 20124, 20240 and 20257 far exceeds that of the other classes due to their high distribution density and low sampling difficulty. Meanwhile, it can be seen from Figure 2 that the presence of many small pests leads to an imbalance in the distribution of the label box size as well. These two types of imbalance problems can interfere with the prior box generation process, resulting in some classes and sizes of pest label boxes dominating the clustering process and reducing the overall match degree of the prior boxes.

To suppress the interference of label box imbalance problem, we propose a prior box generation method based on label box oversampling and clustering, named BS-Medians. In the BS-Medians method, we first use Borderline-SMOTE to oversample the minority class label boxes and suppress the interference of majority class label boxes. Label box oversampling is performed on the set of borderline samples, which refer to the minority class label boxes with the proportion of majority class label boxes between 0.5 and 1 in the k-nearest neighbors. The detailed oversampling process is as follows:

(1) Determine the number of label boxes n_i that need to be generated for the minority class c_i. Where $i = 1,2,\dots,m$, and m is the number of minority classes.

(2) Randomly select a label box x_j from the set of borderline samples of the minority class c_i, and randomly select its nearest neighbor label box $\hat{x}$. Where $j = 1,2,3,\dots,n_i$.

(3) Generate a new label box y_j, as shown in Equation 1:

$$y_j = x_j + \alpha \times (\hat{x} - x_j) \tag{1}$$

Where, α is a random number from 0 to 1.

(4) Repeat steps 2 and 3 to generate n_i label boxes.

(5) Repeat all the above steps to generate label boxes for all the minority classes.

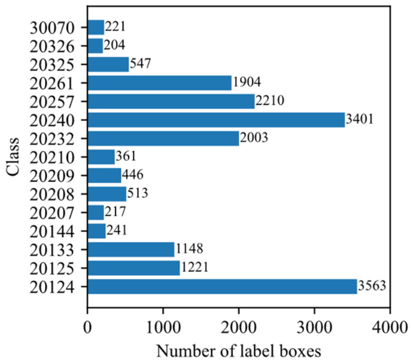

Figure 1. Label box quantity distribution.

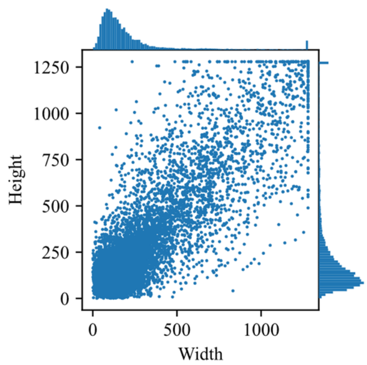

Figure 2. Label box size distribution.

The above label box oversampling process generates many medium and above-sized label boxes for the minority classes, which is beneficial to improve the distribution of the label box quantity among the classes and the distribution of the label box size in the CP15 dataset. Subsequently, BS-Medians uses K-Medians to generate the prior boxes. The K-Medians uses the median to update the clustering centers, which is insensitive to outliers and more suitable for label boxes with imbalance problems.

From the above process, it is clear that BS-Medians method helps to suppress the interference of label box imbalance problem and generate high quality prior boxes. This method is not only applicable to the CP15 crop pest dataset, but also can be used in other datasets with label box imbalance problem to generate prior boxes with high matching degree and improve the precision of crop pest detection.

B. *Optimization of downsampling method*

YOLOLite-CSG contains five strided convolution, which provide multi-scale feature maps for feature fusion. However, the sparse sampling process in the strided convolution is prone to the loss of fine-grained information and leads to the lack of learnable features in the backbone network. Inspired by the downsampling method of SPD-Conv [12], we propose a downsampling method based on parallel feature transformation, PSPD-Conv, to solve the fine-grained information loss problem of YOLOLite-CSG. The overall structure of the PSPD-Conv method is like that of SPD-Conv, but it introduces a parallel feature transformation structure, which is shown in Figure 3. Where Cat denotes connecting feature maps along the depth dimension and the downsampling rate is 2.

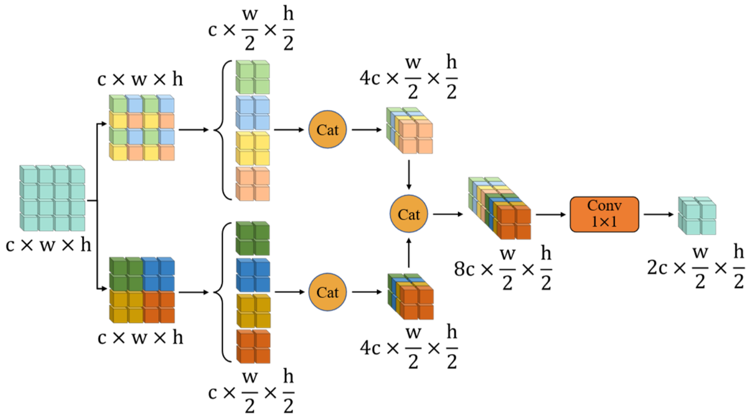

Figure 3. Structure of the PSPD-Conv.

The parallel feature transformation structure is the core of PSPD-Conv downsampling method which is used to segment the feature maps. In this structure, the up branch uses a dilated sampling window to segment the input feature map, and the down branch uses an equal sampling window to segment the input feature map. The feature map segmentation rules for the two branches are shown in Equation 2 and Equation 3:

$$f(i,j) = X(i{:}h{:}s, j{:}w{:}s) \tag{2}$$

$$f(i,j) = X(i \times s{:}i \times s + s{:}1, j \times s{:}j \times s + s{:}1) \tag{3}$$

Where, f denotes the segmented partial feature map, i and j denote the sampling range index, w and h denote the width and height of the input feature map, s denotes the downsampling rate, and X denotes the input feature map.

After feature map segmentation, the partial feature maps are stacked along the depth dimension, preserving all the fine-grained feature information. The feature map formed by the stacking is adjusted in depth by 1×1 convolution to generate the final downsampling result. The spatial resolution and the channel number of this resultant feature map are the same as those of strided convolution downsampling, but more fine-grained feature information is preserved, which is more conducive to improving the crop pest detection precision.

C. Optimization of bounding box regression loss function

The YOLOLite-CSG uses the CIoU [13] loss function to optimize the bounding box regression process, which is shown in Equation 4, Equation 5, and Equation 6:

$$Loss_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{4}$$

$$\alpha = \frac{v}{(1 - IoU) + v} \tag{5}$$

$$v = \frac{4}{\pi^2}\left(arctan\frac{w^{gt}}{h^{gt}} - arctan\frac{w}{h}\right)^2 \tag{6}$$

Where, IoU denotes intersection over union, $\rho^2(b, b^{gt})$ denotes the distance between the center point b of the prediction box and the center point b^{gt} of the label box, and c denotes the diagonal length of the minimum closure. Meanwhile, α denotes the adjustment factor of the aspect ratio difference penalty term, and v denotes the aspect ratio difference penalty term. In addition, w^{gt} and h^{gt} denote the size of the label box, and w and h denote the size of the prediction box.

The CIoU loss function has no penalty term for the size difference, lacks size difference optimization capability, and is prone to function degradation. When the aspect ratios and center points of the prediction box and label box are the same, the penalty terms are both zero and the CIoU loss function degenerates to the inefficient IoU loss function [14]. The function degradation limits the error representation capability of the loss function, which tends to reduce the convergence speed of the bounding box regression process, leading to insufficient convergence of the prediction box.

To address the function degradation, we propose FIoU loss function that incorporates the size difference into the optimization range, as shown in Equation 7, Equation 8, and Equation 9:

$$Loss_{FIoU} = 1 - IoU^\alpha + \left(\frac{\rho^2(b, b^{gt})}{c^2}\right)^\alpha + \left(\frac{p1 + p2}{3}\right)^\alpha \tag{7}$$

$$p1 = \frac{4}{\pi^2}\left(arctan\frac{w^{gt}}{h^{gt}} - arctan\frac{w}{h}\right)^2 \tag{8}$$

$$p2 = \left(\frac{w^{gt} - w}{max(w^{gt}, w)}\right)^2 + \left(\frac{h^{gt} - h}{max(h^{gt}, h)}\right)^2 \tag{9}$$

Where, $p1$ denotes the aspect ratio difference penalty term, $p2$ denotes the size difference penalty term, α denotes the exponential factor, and the rest of the component elements have the same meaning as Equation 4.

As can be seen from Equation 7, even if the aspect ratios and center points of the prediction box and label box are the same, FIoU loss function is still able to optimize the bounding box regression process using the overlap degree loss and the size difference penalty term instead of degenerating into the inefficient IoU loss function. Therefore, FIoU loss function can quickly and completely optimize the bounding box regression process and produce better bounding box regression results than CIoU loss function. In addition, inspired by Alpha-IoU [15], we introduced an exponential factor α for the FIoU loss function. The factor can increase the gradient and loss of high IoU pest objects, suppress the interference of noisy information on the training, and improve the convergence speed and precision of the bounding box regression process.

D. Optimized lightweight crop pest detection method

Based on the above optimization methods, we optimized YOLOLite-CSG to build an optimized lightweight crop pest detection method YOLOLite-X. YOLOLite-X employs BS-Medians to generate high-quality prior boxes, relies on PSPD-Conv to complete image downsampling without fine-grained information loss, and uses the FIoU loss function to solve the function degradation problem and optimize the bounding box regression process. The structure of YOLOLite-X is shown in Figure 4. Where, Conv denotes the convolutional layer, PSPD BCA denotes the combination of PSPD-Conv downsampling method and Bottleneck-CA, which is a residual block containing coordinate attention, and PSPD LSCA denotes the combination of PSPD-Conv downsampling method and LightSandGlass-CA, which is a lightweight sandglass block containing coordinate attention.

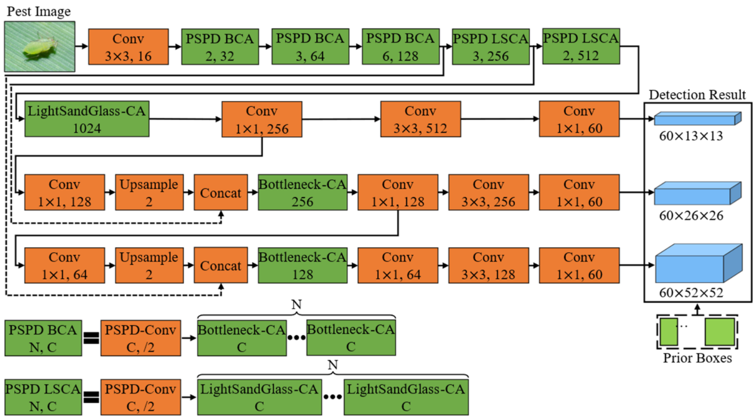

Figure 4. Structure of YOLOLite-X.

III. Experiment results analysis and discussion

A. CP15 dataset

All images of the original dataset were collected from agricultural environments. Since some of the original samples had duplicates and large label errors, we recalibrated the original labels and removed images with no pests or pest objects that were too ambiguous. Meanwhile, we used data augmentation methods such as horizontal flip and Cutout to expand the original dataset. In addition, since the original images are too high resolution for direct input to the neural network, we crop each pest image into several images according to the preset sampling window size and sampling stride. After the above processing, we built the CP15 crop pest dataset.

B. Ablation experiment

We conducted ablation experiment based on the CP15 dataset to evaluate the precision, number of parameters and computations of the crop pest detection method. According to the results of the ablation experiment in Table 1, the number of parameters and computations of the crop pest detection method remain unchanged after the introduction of BS-Medians prior box generation method, and the mAP is improved by 0.3%. This improvement proves that BS-Medians method can generate high-quality prior boxes and improve the precision of crop pest detection. Subsequently, PSPD-Conv method was used for pest image downsampling, and the number of parameters and computations of the crop pest detection method decreased slightly, and the mAP increased by 0.7%. The result shows that PSPD-Conv downsampling method preserves more fine-grained feature information and improves the usability of texture, edge and other pest features. Finally, FIoU loss function improves the convergence degree and speed of the bounding box regression using its overall error representation capability, resulting in a 0.9% increase in the mAP of the crop pest detection method. With the above optimization methods, the optimized lightweight crop pest detection method YOLOLite-X is built.

Table 1. Results of the ablation experiment.

Method	mAP (%)	Params (M)	FLOPs (G)
YOLOLite-CSG	82.9	5	9.8
+ BS-Medians	83.2	5	9.8
+ PSPD-Conv	83.9	4.8	9.6
+ FIoU Loss (YOLOLite-X)	84.8	4.8	9.6

C. Comparison experiment

We conducted the comparison experiment based on the CP15 dataset and evaluated the mAP, number of parameters and computations of YOLOLite-X and YOLOv3 and other methods. According to the experiment results in Table 2, the crop pest detection precision of YOLOLite-X was significantly improved compared with YOLOLite-CSG, and the mAP increased by 1.9%. Moreover, the number of parameters and computations of YOLOLite-X decreased slightly, which has obvious comprehensive performance advantages. In addition, the crop pest detection precision of YOLOLite-X is significantly higher than that of other state-of-the-art object detection methods such as YOLOv5m and EfficientDet-D2, with mAP leading by more than 1%. YOLOLite-X also has a significant advantage in terms of number of parameters and computations, which is only equivalent to lightweight object detection methods such as NanoDet. Therefore, YOLOLite-X has high availability to meet the needs of crop pest control work in agricultural environments that lack high-performance devices.

Table 2. Results of the comparison experiment.

Method	mAP(%)	Params(M)	FLOPs(G)
Faster R-CNN	73.16	137	253
SSD	75.5	25.5	118.1
RetinaNet	78.92	36.6	71
YOLOv3-Tiny	72.3	8.7	5.5
YOLOv3	82.4	61.6	65.5
YOLOv5s	78.7	7.1	6.7
YOLOv5m	82.6	20.9	20.4
CenterNet	73.03	32.7	46.2
FCOS	76.88	32.1	68.2
NanoDet-Plus	78.8	7.8	3
EfficientDet-D2	83.14	8	20.5
YOLOLite-CSG	82.9	5	9.8
YOLOLite-X	84.8	4.8	9.6

IV. Conclusion

In this paper, we analyze YOLOLite-CSG, and propose various optimization methods to build an optimized lightweight crop pest detection method with higher precision, named YOLOLite-X. To optimize YOLOLite-CSG, we propose BS-Medians prior box generation method, PSPD-Conv downsampling method, and FIoU bounding box regression loss function. The BS-Medians method improves the balance of label box distribution to increase the matching degree and size range of prior boxes, the PSPD-Conv downsampling method solves the problem of fine-grained feature information loss to improve the detection precision of crop pests, and the FIoU bounding box regression loss function solves the function degradation problem to improve the convergence speed of YOLOLite-X. After the above optimizations, the detection precision of YOLOLite-X reaches 84.8%, which is higher than that of YOLOLite-CSG. Meanwhile, the number of parameters and computations of YOLOLite-X are less than those of YOLOLite-CSG, and the practical usability is stronger.

Acknowledgments

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